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Praveen-Kumar-Energy

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✎

Dr. Polamarasetty Praveen Kumar

Energy Asset Optimisation · Multi-Market Bidding · MILP Dispatch · Risk Analytics

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📊 H-INDEX 22

📅 AVAILABLE

📅 JULY 2026

📄 EU WORK AUTHORIZATION

🇵🇹 PORTUGAL

📄 PUBLICATIONS

📈 70+ | 1,500+ CITATIONS

About Me

Energy systems researcher and industrial-grade software engineer with deep expertise in hybrid plant optimisation, multi-market bidding ar scheduling, and MILP-based dispatch.

Built a 14-gate multi-market MILP bidding and scheduling system (DA + IDA1/2/3 + XBID + aFRR + mFRR + ISP dispatch + settlement + analytics) for the real Alqueva hybrid plant (518.4 MW PSP + 5 MW PV + 1 MW / 2 MWh BESS, Portugal / MIBEL). System uses IBM CPLEX (av solve 0.172 s, mip_gap=0.005), auto-selects ML forecasters (Naive / Ridge / LightGBM via walk-forward CV), and applies a permanent 13-invariant physical checker that blocks any bid with a constraint violation before it propagates. Risk framework includes historical VaR/CVaR (9 % & 99 %) and Monte Carlo bootstrap confidence intervals (n=10,000).

⚡ Flagship Project — Alqueva 24-Hour Energy Trading Optimizer

🐍 PYTHON

📦 3.10+

🔧 SOLVER

📦 CPLEX 22.1

🔄 FALLBACK

📦 HIGHS | CBC

🔧 OPTIMISATION

📦 MILP

🌐 MARKETS

📄 DA | IDA1-3 | XBID | AFRR | MFRR

🌐 GRID

📄 MIBEL / OMIE / REN

🔧 PIPELINE

📈 15 PHASES | 125S

Plant

Asset	Specification
PSP — Pumped Storage	4 × 129.6 MW reversible Francis units · 518.4 MW generation / 446.4 MW pumping
PV — Floating Solar	5 MWp · commissioned 2022 · temperature derate + annual degradation model
BESS — Battery	1 MW / 2 MWh · SOC 10–95 % · $\eta_{\text{charge}} = \eta_{\text{discharge}} = 0.90$
Upper Reservoir	Alqueva · 3,150 hm³ usable · head range 54.7–73.0 m
Lower Reservoir	Pedrógão · 54 hm³ usable · binding constraint on long pumping sequences

15-Phase Pipeline

Phase	Gate	Key Output
1 — DA	OMIE D-1 12:00	H1–H24 energy bids · +€127,767 revenue
2A — IDA1	OMIE SIDC D-1 15:00	Position delta re-optimisation · H1–H24
2B — IDA2	OMIE SIDC D-1 22:00	Position delta · H3–H24
2C — IDA3	OMIE SIDC D 10:00	Position delta · H12–H24 (H1–H11 frozen)
2D — XBID	SIDC continuous H-1	Cross-border continuous intraday
3A — aFRR	PICASSO DA+1h	Symmetric ±MW capacity · FAT 5 min · +€103,324
3B — mFRR	MARI DA+1h	Symmetric ±MW capacity · FAT 12.5 min · +€23,518
4A — RT Dispatch	96 ISPs × 15 min	REN signal simulation
4B — aFRR Activation	Real-time	Activation response
4C — mFRR Activation	Real-time	Activation response
5A — Energy Settlement	Post-delivery	DA / IDA imbalance reconciliation
5B — Reserve Settlement	Post-delivery	aFRR / mFRR payment
5C — Imbalance Settlement	Post-delivery	REN 15-min ISP penalty calculation
5D — Analytics	End-of-day	KPI report · 5-sheet Excel · 9 figures

```
$ python run_production.py --auto --synthetic
```



PHASE		STATUS	TIME	NOTE
1	Day-Ahead bidding (OMIE DA)	[OK]	50.65s	energy revenue +127,767
2A	IDA1 intraday re-optimisation	[OK]	10.78s	
2B	IDA2 intraday re-optimisation	[OK]	11.79s	
2C	IDA3 intraday re-optimisation	[OK]	10.70s	
2D/w1	XBID continuous (D-1 18:30)	[OK]	9.96s	
3A	aFRR capacity offer (PICASSO/REN)	[OK]	10.83s	capacity revenue +103,324
3B	mFRR capacity offer (MARI)	[OK]	3.97s	capacity revenue +23,518
4A	RT dispatch simulation (96 ISPs)	[OK]	0.03s	
4B	aFRR activation response	[OK]	0.21s	
4C	mFRR activation response	[OK]	0.13s	
5A	Energy settlement (DA / IDA)	[OK]	1.89s	
5B	Reserve settlement (aFRR / mFRR)	[OK]	0.04s	
5C	Imbalance settlement (REN balance)	[OK]	0.18s	
5D	Analytics + KPI report + Excel	[OK]	2.15s	total pnl +312,886

PIPELINE COMPLETE — 15 passed 0 failed (125.5s total)

Architecture Highlights

- **Single shared MILP model** — `core_milp_solver.py` used by all 15 phases; each phase passes its own horizon, frozen variables, and mark constraints. IBM CPLEX avg solve **0.172 s**, `mip_gap` = 0.005
- **--from-phase hot-restart** — re-enter at any phase without re-running earlier phases; production-safe for intraday gate re-optimisation
- **ComponentStore** — typed dataclass store (GateResults, DispatchPlan, SettlementSheet) shared across phases; zero file I/O between phas
- **13-invariant physical checker** — re-derives dispatch from solved schedule; checks mode exclusivity, min stable load, ramp rates, reservoir bounds, BESS SoC, energy balance, no-double-sell, FAT deliverability; raises `BidCheckError` on any violation before propagation
- **ML forecasting pipeline** — DA price, aFRR, mFRR, PV power, water inflow forecasters each auto-select best model (Naive / Ridge / LightGBM) via 4-fold walk-forward CV; 17 engineered features for DA price (lag 24h/48h/168h/336h, rolling mean/std, cyclical calendar)
- **Risk analytics** — historical VaR (95%/99%) and CVaR/Expected Shortfall; Monte Carlo bootstrap (n=10,000 resamples); Sharpe ratio (annualised); max drawdown; P&L attribution by stream (DA / IDA / aFRR / mFRR / imbalance)
- **Immutable audit trail** — append-only JSON-lines log of every decision (solve / check / approve / submit) with market-time timestamps; designed for regulatory inspection
- **CPLEX 22.1 primary / HiGHS fallback** — automatic solver detection; same model runs on any machine
- **15-min ISP resolution** — 96 intervals/day matching REN grid operator standard (Mar 2025 regulatory)

Daily P&L Summary

Revenue Stream	Amount
DA energy trading (OMIE)	+€127,767
aFRR capacity (PICASSO)	+€103,324
mFRR capacity (MARI)	+€23,518
IDA re-optimisation + XBID	included
Imbalance settlement	included
Total P&L	+€312,886

 Technical Skills

Area	Detail
Optimisation	MILP formulation · Pyomo · IBM CPLEX (mip_gap=0.005, avg 0.172 s) · HiGHS · CBC · unit commitment · piecewise-linear turbine/pump curves · McCormick linearization · 24-hour scheduling horizon
Energy Markets	OMIE DA/IDA · SIDC IDA1/2/3 (post-Jun 2024 reform) · aFRR via PICASSO (FAT 5 min) · mFRR via MARI (FAT 12.5 min, live 27 Nov 2024) · XBID continuous intraday · ISP imbalance settlement (15-min from 19 Mar 2025) · FCR non-remunerated headroom reservation
Risk & P&L	VaR & CVaR — historical simulation + Monte Carlo bootstrap (n=10,000) · confidence intervals · Sharpe ratio (annualised) · max drawdown · P&L attribution by stream · pre-trade risk checks · audit logging
ML & Forecasting	LightGBM · Ridge regression · Naive baseline · walk-forward CV (4 folds) · lag/rolling/cyclical calendar features · DA price, aFRR, mFRR, PV power, inflow forecasters
Programming	Python · pandas · NumPy · Pyomo · scikit-learn · LightGBM · openpyxl · SQLite · Git/GitHub · YAML config · CLI design · pytest
Systems & Data	OMIE live API · PICASSO/MARI/REN structured data · ENTSO-E market frameworks · ERA5 reanalysis (PV weather) · SQLite position/reserve/delivery/audit stores · Excel report generation
Asset Knowledge	Pumped-storage hydropower (Francis reversible units) · floating solar PV · BESS (SoC tracking, cycle degradation) · reservoir dynamics (upper/lower bounds, natural inflow) · head-dependent efficiency curves

 Professional Experience

Energy Optimisation Engineer (PostDoc Researcher)

INESC TEC — Centre for Power and Energy Systems (CPES), Porto, Portugal · Feb 2025 – Present

STOR-HY (Horizon Europe GA 101172905) · Alqueva Hybrid Plant: 4 × 129.6 MW reversible Francis PSP + 5 MWp Floating PV + 1 MW / 2 MWh BESS

- **Built a 14-gate end-to-end automated bidding and scheduling pipeline** covering DA bid (OMIE), IDA1/2/3 re-optimisation (SIDC), XBID continuous intraday, aFRR capacity offer (PICASSO), mFRR capacity offer (MARI), ISP dispatch simulation (REN signal architecture), aFRR/mFRR TSO activation, DA/IDA energy settlement, reserve settlement, imbalance settlement, and daily P&L analytics — all orchestrated from a single `run_production.py` with full audit logging; IDA bids are position deltas (post-June-2024 MIBEL reform); IDA3 restricted to hours 12–24; CPLEX avg solve 0.172 s
- **ML forecasting pipeline:** DA price, aFRR price, mFRR price, PV power, and water inflow forecasters each auto-select the best model (Naive / Ridge / LightGBM) via 4-fold walk-forward cross-validation; 17 engineered features for DA price (lag 24h/48h/168h/336h, rolling mean/std, cyclical calendar encodings)
- **Physical plant modelling:** piecewise-linear turbine efficiency curves (4 Francis units, 129.6 MW each); pump flow linearized for MILP compatibility; full upper (Alqueva 3,150 hm³) and lower (Pedrógão 54 hm³) reservoir dynamics with monthly natural inflow; BESS SoC tracking with binary charge/discharge commitment and cycle-weighted degradation cost
- **Permanent 13-invariant physical checker:** re-derives dispatch from solved schedule and replays through plant models before any result propagates — checks mode exclusivity, min stable load, ramp rates, reservoir bounds, BESS SoC, energy balance, no-double-sell, and FAT deliverability; raises `BidCheckError` on any violation
- **Risk analytics framework:** historical VaR (95%/99%) and CVaR/Expected Shortfall from daily P&L distribution; Monte Carlo bootstrap (n=10,000 resamples with replacement) computes confidence intervals on VaR and CVaR; annualised Sharpe ratio and max drawdown; all metrics exported to Excel Risk sheet

- **Immutable audit trail:** append-only JSON-lines logging every decision (solve/check/approve/submit) with market-time timestamps; designed for regulatory inspection and post-trade accountability
- Participated in consortium technical meetings with **EDP Produção**, **EDP New Energy Technologies**, **EDF**, and **GE Vernova** within the **STOR HY Horizon Europe** project

Assistant Professor

GMR Institute of Technology, Rajam, India — Dept. Electrical & Electronics Engineering · Jul 2021 – Jan 2025

Power systems and renewable energy teaching; hybrid energy systems research.

PhD Research Scholar

IIT Roorkee — Dept. of Hydro and Renewable Energy · Jul 2018 – Jul 2021

- Designed and optimized hybrid renewable energy systems (PV, wind, biomass, BESS) for rural electrification in India
- Applied metaheuristic optimization in MATLAB to minimize LCC, LCOE across multiple battery technologies and dispatch strategies
- Published complete optimization code on MathWorks File Exchange (open-source, reproducible research)

Education

Degree	Institution	Year
Ph.D. — Renewable Energy	IIT Roorkee, India	2018 – 2021
M.E. — Power Electronics Drives & Control	Andhra University, India	2010 – 2012
B.Tech — Electrical & Electronics Engineering	JNTU Kakinada, India	2006 – 2010

Industrial Collaborations & Funded Projects

STOR-HY · Horizon Europe GA 101172905 · 2025–Present
Partners: EDP Produção · EDP New Energy Technologies · EDF · GE Vernova
Built core MILP optimisation system for the Alqueva hybrid plant

SE-HYDRO · Horizon Europe GA 101269565 · Started May 2026
16-partner consortium: NTUA (coordinator) · ANDRITZ HYDRO · EDP · GE Vernova · PPC Greece · Smart Innovation Norway
Co-authored successful Horizon Europe grant proposal; contributing to AI-based energy management and digital twin frameworks for hydro pilot sites in Greece, France, Portugal and Serbia

Research Impact

PUBLICATIONS	70+	H-INDEX	22	CITATIONS	1,500+
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Contributors 1



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